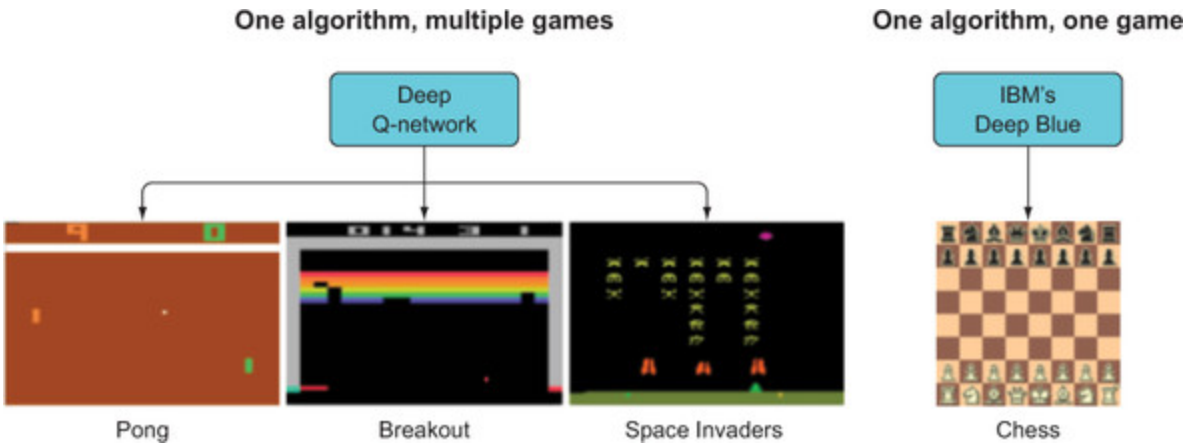


More recently, DeepMind's AlphaGo and AlphaZero algorithms beat the world's best players at the ancient Chinese game Go. Experts believed artificial intelligence would not be able to play Go competitively for at least another decade because the game has characteristics that algorithms typically don't handle well. Players do not know the best move to make at any given turn and only receive feedback for their actions at the end of the game. Many high-level players saw themselves as artists rather than calculating strategists and described winning moves as being beautiful or elegant. With over 10^{170} legal board positions, brute force algorithms (which IBM's Deep Blue used to win at chess) were not feasible. AlphaGo managed this feat largely by playing simulated games of Go millions of times and learning which actions maximized the rewards of playing the game well. Similar to the Atari case, AlphaGo only had access to the same information a human player would: where the pieces were on the board.

While algorithms that can play games better than humans are remarkable, the promise and potential of RL goes far beyond making better game bots. DeepMind was able to create a model to decrease Google's data center cooling costs by 40%, something we explored earlier in this chapter as an example. Autonomous vehicles use RL to learn which series of actions (accelerating, turning, breaking, signaling) leads to passengers reaching their destinations on time and to learn how to avoid accidents. And researchers are training robots to complete tasks, such as learning to run, without explicitly programming complex motor skills.